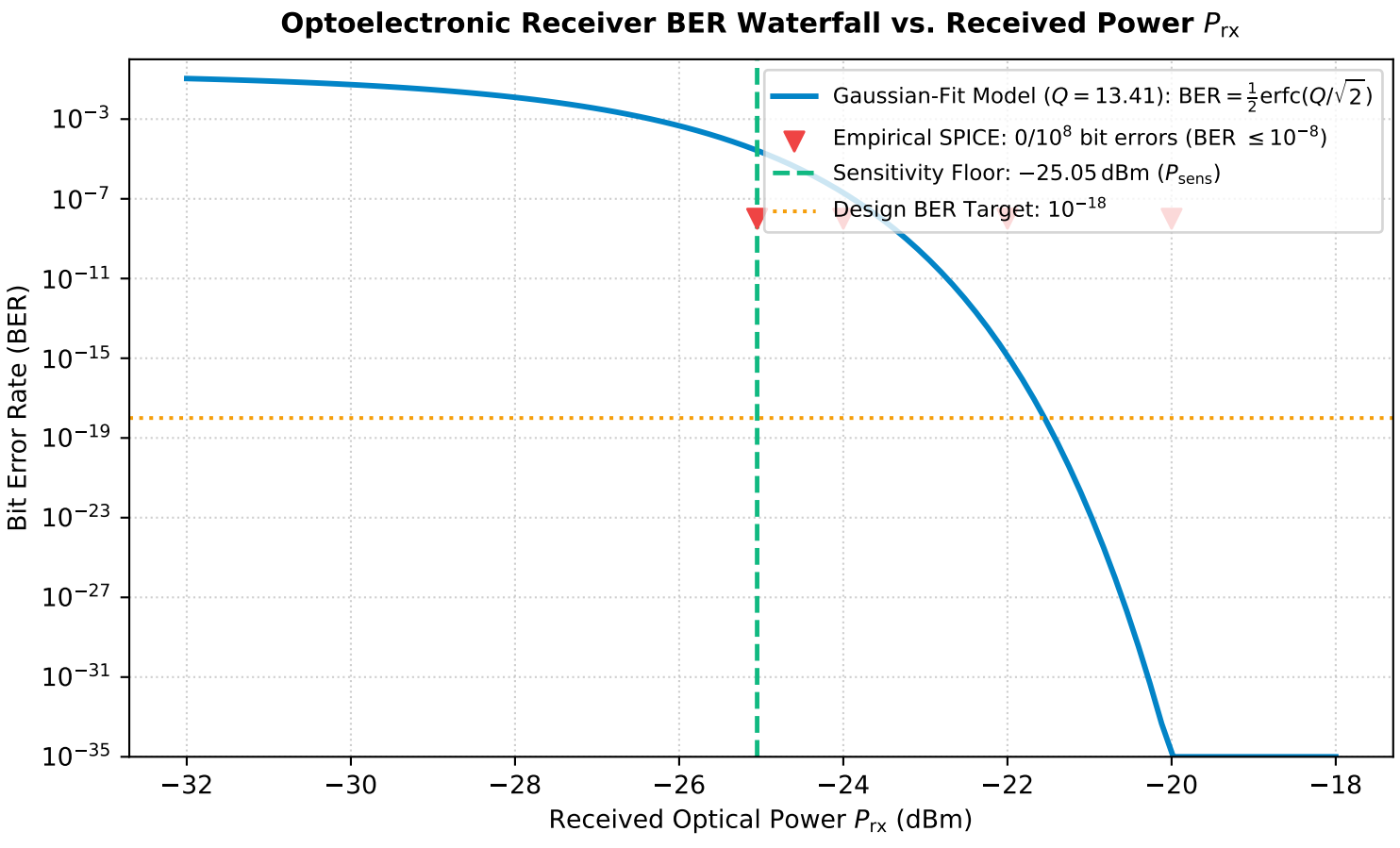
Optoelectronic Receiver BER Waterfall vs. Received Power $P_{rx}$
Gaussian-Fit Model ($Q = 13.41$): BER $= \frac{1}{2}\mathrm{erfc}(Q/\sqrt{2})$
Empirical SPICE: $0/10^8$ bit errors (BER $\leq 10^{-8}$)
Sensitivity Floor: $-25.05$ dBm ($P_{sens}$)
Design BER Target: $10^{-18}$
Bit Error Rate (BER)
$10^{-3}$
$10^{-7}$
$10^{-11}$
$10^{-15}$
$10^{-19}$
$10^{-23}$
$10^{-27}$
$10^{-31}$
$10^{-35}$
−32
−30
−28
−26
−24
−22
−20
−18
Received Optical Power $P_{rx}$ (dBm)